Microcontrollers Reflection

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The *Microcontroller and Circuits* Module introduced us to programming as a whole and then narrowed our focus on to a specific microcontroller, an Arduino Nano Esp32, and its language, Micro Python. We got have a hands-on experience on circuit building through the use of breadboards. Through it, we implemented common elements found in circuits such as LED's, potentiometers, and resistors. The module introduced us to common conventions in circuit building such as associating black with ground (GND), and red with power (POW).

A large component of the module was just familiarizing us with the iterative process by giving us new additions on top of what we'd already created. For example, the very first circuit we built was a simple circuit with a potentiometer and an LED. We were then tasked with replacing out potentiometer with switch, a button, and using it to turn on and off the Led. After which we once replaced the Switch with a light sensor.

This overall process helped to solidify the concept of transitioning from manual to automated. The first two circuits involved a person's involvement, however the final circuit had light act as an "automated" trigger. Engineering often involves optimization, such as making a skyscraper that's as high as possible, but only having a certain area to work with. While, this simplistic circuit is a long shot from my example, exposing us to this concept in a variety of ways helps to reinforce it and create a mindset that'll make us successful engineers in the future.

The last portion of this part of the Module was using everything that'd we learned up to this point in order to create an intruder alarm system with the given materials. My group began by seeing what components of the alarm system could be replaced by what we had. For example, we used button as the switch for the Activate portion of the system diagram and the LED as our Alarm in our system diagram. From here we transitioned to the flow chart of our prototype where we distilled down what our future code would do. Our chart consisted of the Arduino Reading the Photoresistor and Button and from there turning the LED on (Armed), if both the switch was pressed and light sensor gave a low reading.

With this plan in place, we used ChatGPT in an LLM assisted approach to develop our code. Here we ran into several problems as we quickly found out, without context, ChatGPT's output is very general. ChatGPT and other LLM's outputs are only as good as the prompts you give it terms of specificity and clarity. The first round of code it generated wasn't usable because it didn't specify which components were connected to which GPIO

Pins. We had to give it the information and even then, it didn't work, because of this we had to go through the debugging process.

Here, I learned the importance of being able to understand what the code is doing and knowing the functions that would be associated with the task at hand. What we found useful as a group was breaking down the code into chunks and seeing whether the code worked plus whether it did its intended purpose. While tedious, we got down to the root of the problem, an OR statement was used instead of an AND statement. Thanks to being able to understand the actual code, we were able to manually rewrite this small snippet of code ourselves rather than asking ChatGPT to do if. We learned that if you explain clearly what you want, ChatGPT misunderstands and leads to more problems down the line because it's generating code that won't ever do what you want it to do.

The latter half of the module built on what skills we'd developed during the first half and introduced us to PWM (Pulse Width Modulation) Square Waves and their role in servomotors. Essentially, the length of the square wave dictates the angular position of the servomotor. Being given this, we were asked to have the servo motor to move to one position when a switch was pressed and another when pressed again. We began by breaking it down into a system diagram just to get an overall view of all the components and separating inputs from outputs. Once again, we used an LLM assisted approach with ChatGPT. Using the lessons from the previous module we got the code to work much quicker with a minimal number of prompts. Being given a simple implementation of a servomotor will make its future potential use in our final project smoother and easier to implement.

We then got rid of the servomotor and incorporated a rain drop sensor with a DC Power Supply and a Digital Multimeter. After setting them to their appropriate settings, we examined the rain drop sensor's voltage under different conditions of "wetness." We saw that there was a ton of variability in the readings and that it was hard to pinpoint exactly what the voltage reading was. In our group, we each looked at the Multimeter and came to a conclusion as a group. This manual method of collecting voltage was inconsistent and didn't have a ton of accuracy. Therefore, the next step of the lab was to use our Arduino and automate the process by having it read the ADC values. This time ChatGPT was not the sole reason our code didn't work. We didn't have the circuit wired as it was supposed to so the Arduino was only reading the ground voltage. Since we understood the code and once we got it do what it was supposed we were able to determine that it was a hardware problem.

Once we got it working, we repeated the previous conditions, having it read 10 times, converting it to volts, and exporting the data as CSV Files. We used this for the last portion of the module: Sensor Performance Analysis. Here we used Google Sheets and

functions such as AVERAGE and STD DEV for both the Automated and Manual measurements. Lastly, incorporating all our data into a chart and fitting it with an appropriate trendline, which in our case was linear. This overall process helped to show how useful Google Sheets/Excel are in performing operations with large data sets. The ability to call cells in other tabs is extremely useful in terms of dividing data in order to keep the overall sheet organized and legible. This skill and many others like this will set us up for success was throughout the rest of our college careers and extend into our professional careers as engineers.